

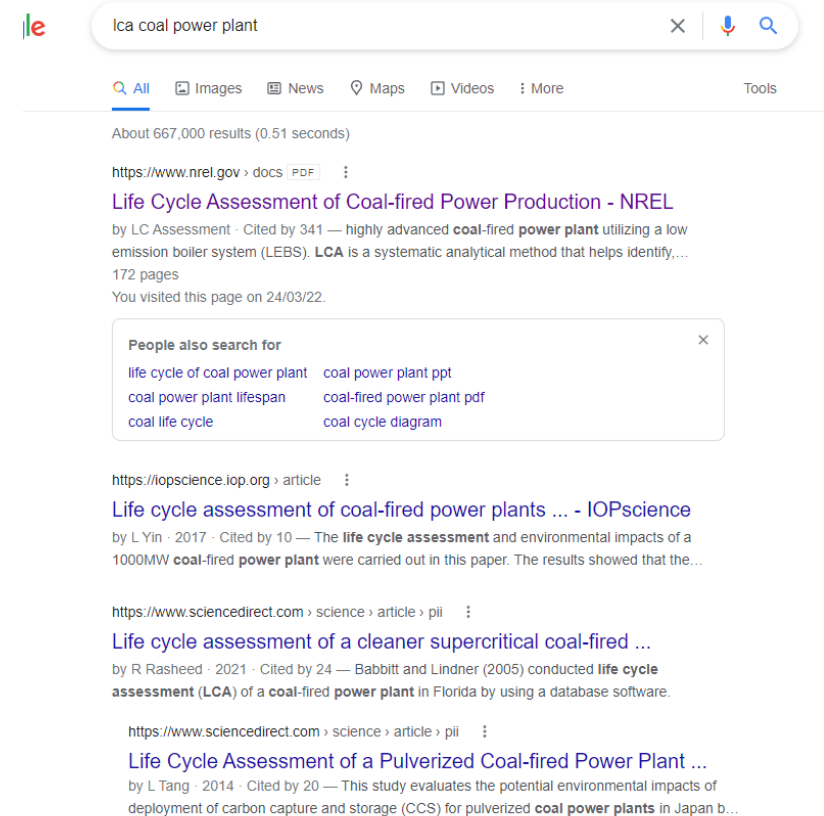
LCA 5 – Literature Review

ENGEN370

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Literature Review Tips

- Google search – “LCA coal power plant”
- Pick the one with the most citations first
- Skim through and look at the tables and graphs
- Look through the references (working backwards)
- Look who cited the paper (working forward)
- Don’t spend too much time, probably 5 references is enough to get started
- Have enough information from 2 references to do a basic LCA around a coal fired power plant of any size (assuming linear scaling)



Spath, Mann, and Kerr 1999, Life Cycle Assessment of Coal-fired Power Production

Materials for construction

Table 42: Plant Material Requirements Used in Sensitivity Analysis

Material	Base case (kg/MW plant capacity)	Sensitivity - case D (kg/MW plant capacity)
Concrete	158,758	6,962
Steel	50,721	1,764
Aluminum	419	310
Iron	619	622

Resources used

Table 37: Resource Consumption

Resource	Average (a, b)		NSPS (a, b)		LEBS (a, b)	
	% by wt (c)	g/kWh (d)	% by wt (c)	g/kWh (d)	% by wt (c)	g/kWh (d)
Coal	80.4	475.62	78.0	434.92	97.3	353.37
Limestone	17.4	102.84	19.7	109.49	0.0	0.04
Oil	1.9	11.48	2.0	11.32	1.3	4.88
Natural gas	0.2	1.25	0.2	1.26	1.3	4.53
Iron ore	0.0	0.11	0.0	0.11	0.0	0.095
Iron scrap	0.0	0.12	0.0	0.12	0.0	0.10

- (a) Results are reported for the surface mining case. However, the underground mining numbers are listed here.
- (b) Coal transportation = average user by river.
- (c) Percent of total resource consumption. Not all resources consumed by the system are shown; they do not add up to 100%.
- (d) Resource consumption per kWh of net electricity produced averaged over the life of the system.

Emissions

Table 29: CO₂ Results

System	Total CO ₂ emissions (g/kWh) (a)	CO ₂ from surface mining subsystem (b)		CO ₂ from transportation subsystem (c)		CO ₂ from power generation subsystem	
		(g/kWh) (a)	(%)	(g/kWh) (a)	(%)	(g/kWh) (a)	(%)
Average	1,022	9	0.9	17	1.7	996	97.3
NSPS	941	8	0.9	16	1.7	917	97.4
LEBS	741	7	1.0	13	1.8	721	97.3

- (a) CO₂ emissions per kWh of net electricity produced averaged over the life of the system.
- (b) Results are reported for the surface mining case. However, the underground mining numbers are similar to those listed here.
- (c) Coal transportation = average user by river.

Wastes

Table 54: Power Plant FGC Waste and Ash

Solid waste	Emission amount (g/kWh) (a)		
	Average	NSPS	LEBS
FGC waste - dry total	86.0 (b)	91.5 (b)	34.2 (c)
FGC waste - dry landfilled	64.7	68.9	0
Ash - moisture free total	35.7 (d)	32.6 (d)	26.6 (d)
Ash - moisture free landfilled	25.7	23.5	19.1
Total waste produced from the system (mining, transportation, and power generation subsystems) (e), (f), (g)	110.4	113.4	20.7

Yin, Liao, Zhou, Wang, and Ma 2017, Life cycle assessment of coal-fired power plants and sensitivity analysis of CO2 emissions from power generation side

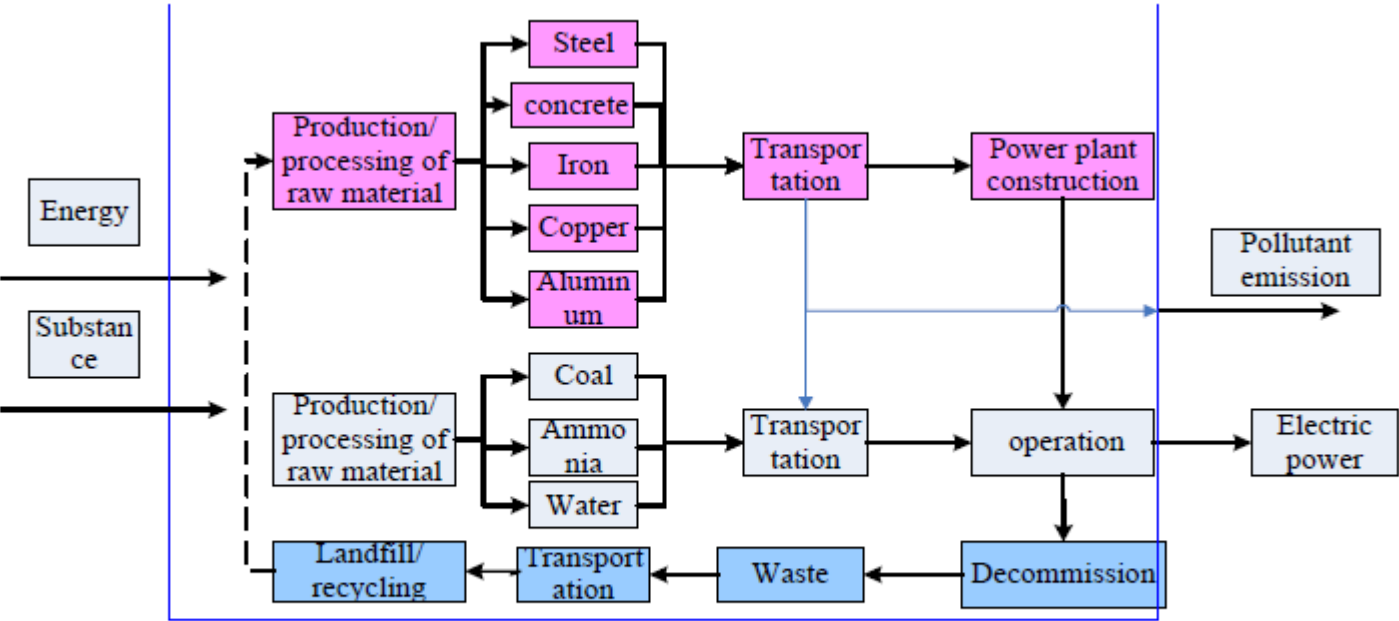


Figure 1. System boundary

Funny units: energy consumption (kg coal/tonne material processed and transported)

Look for conversions: 6.52 kWh = 2.1842 kg coal

Energy consumption and emissions for mining, smelting, and transporting raw materials in China					
Conversion	2.1842	kg coal	=	6.52	kWh
		Energy consumed (kg coal/tonne)		Energy (kWh/tonne)	CO2 (kg/tonne)
		Material			
		Steel	1524	4548	2436
		Concrete	192	572	689
		Fe	536	1599	1536
		Cu	120	359	18382
		Al	78	232	822

Table 1. The energy consumption required for mining, smelting and transporting of raw materials and gaseous pollutants (kg/t)

Item	Standard coal	CO ₂	SO ₂	NO _x	CH ₄	CO	VOC	N ₂ O	COD	Dust
Steel	1523.62	2435.862	5.914794	5.242415	0.009022	0.80976	0.178762	0.004041	0.254	93.42482
Concrete	191.6065	689.364	0.813211	1.37852	0.896079	0.12581	—	—	0.242358	1.031
Fe	535.73	1535.862	4.034742	5.242415	0.009022164	0.809759883	0.178762	0.00404137	—	32.72135
Cu	120.1	18381.8	1298.1	54.1						
Al	77.8	822	2.09	2.98	9.16	0.761	—	—	—	2.39

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[CT Hendrickson](#), [LB Lave](#), [HS Matthews](#) - 2010 - [taylorfrancis.com](#)

Environmental life cycle assessment is often thought of as cradle to grave and therefore as the most complete accounting of the environmental costs and benefits of a product or ...

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[HTML] Life cycle energy and environmental performance of a new university building: modeling challenges and design implications

[C Scheuer](#), [GA Keoleian](#), [P Reppe](#) - [Energy and buildings](#), 2003 - Elsevier

A comprehensive case study life cycle assessment (LCA) was conducted of a 7300 m², six-story building with a projected 75 year life span, located on the University of Michigan ...

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Full cost accounting for the life cycle of coal

[PR Epstein](#), [JJ Buonocore](#), [K Eckerle](#)... - [Annals of the New York Academy of Sciences](#), 2011 - [Wiley Online Library](#)

Each stage in the life cycle of coal—extraction, transport, processing, and combustion—generates a waste stream and carries multiple hazards for health and the environment ...

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[HTML] Environmental impact and life cycle assessment (LCA) of traditional and 'green'concretes: Literature review and theoretical calculations

[P Van den Heede](#), [N De Belie](#) - [Cement and Concrete Composites](#), 2012 - Elsevier

With the current focus on sustainability, it is necessary to evaluate concrete's environmental impact properly, especially when developing new 'green'concrete types. Therefore, we ...

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Life cycle assessment of hydrogen production via natural gas steam reforming

[PL Spath](#), [MK Mann](#) - 2000 - [osti.gov](#)

A life cycle assessment of hydrogen production via natural gas steam reforming was performed to examine the net emissions of greenhouse gases as well as other major ...

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Renewable energy from willow biomass crops: life cycle energy, environmental and economic performance

[GA Keoleian](#), [TA Volk](#) - [BPTS](#), 2005 - [Taylor & Francis](#)

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LCA around concretes

LCA around biomass

Evaluating the information

- Context – is it for a similar or same application?
- Location – is it for a similar or different geographical location?
 - E.g. Yin 2017 paper, Table 1 is for materials made in China – where are you getting your materials from?
 - Metals – probably China
 - Concrete – New Zealand
 - Aluminium – New Zealand – but bauxite mined in Australia
 - Is it a similar process for manufacture?
 - Aluminium made in NZ will have a lower environmental impact
 - Concrete manufacture will depend on the kiln and fuel used for making the calcium oxide from limestone
 - Emissions from energy use will depend on the country and its energy mix

Evaluating the information

- Time – is the data from within the last 5-10 years?
 - Has the method of manufacture changed
 - Has there been any improvements in efficiency of manufacture
- Method of calculating the data
 - Actual measurements (high quality)
 - Based on databases (good/high quality)
 - Based on other journal papers (ok/good quality)
 - Based on calculations (ok quality)
 - Based on assumptions (low quality)
 - Based on opinion, back of the packet calculations (very low quality)

Reporting the paper in your LCA

- Keep it concise
 - E.g. Spath, Mann, and Kerr (1999) carried out an LCA around three types of coal fired power generation and found that 97% of the CO₂ emissions was from electricity generation over the lifetime of the plants.
- Include any key tables/figures from the paper and reference those tables/figures in your text
 - E.g. Spath, Mann, and Kerr (1999) found that concrete was the main material used in the construction of coal fired power plants (Table 1 or whatever table number it is).
- Include some critique if needed
 - E.g. Spath, Mann, and Kerr (1999)'s LCA was based on measured plant data from the 1990's, therefore the data may be out of date.

Activity

- Use Google to find 2 or more papers related to your LCA
- Choose one and find a relevant table that will be useful for your LCA
- Looking through the references, find a paper that will be relevant to your LCA
- Choose one and use Google Scholar to find who has cited your paper, and pick one that is relevant to your LCA
- Choose one parameter from the paper and evaluate the quality of that parameter